

III.

$$\begin{aligned}
 (a) \mathbb{E}[y^T W y] &= \mathbb{E}[\text{tr}(y^T W y)I] \\
 &= \mathbb{E}[\text{tr}(W y y^T)] \\
 &= \text{tr}(\mathbb{E}[W y y^T]) \\
 &= \text{tr}(W(\Sigma + \mu \mu^T)) \\
 &= \sigma^2 \text{tr}(W) + \text{tr}(\mu^T W \mu) \\
 &= \sigma^2 \text{tr}(W) + \beta^T X^T W X \beta
 \end{aligned}$$

(b) $C\hat{\beta} = C(X^T X)^{-1} X^T y$; normal

$$\mathbb{E}[C\hat{\beta}] = C\beta$$

$$\text{var}(C\hat{\beta}) = \sigma^2 C(X^T X)^{-1} C^T$$

$$\therefore C\hat{\beta} \sim N(C\beta, \sigma^2 C(X^T X)^{-1} C^T)$$

(c) $\hat{\beta}_c = (I - B)\hat{\beta}$

$$\text{cov}(\hat{\beta}_c, C\hat{\beta}) = (I - B)(X^T X)^{-1} C^T = 0$$

(d) $\sigma^2(I - B)(X^T X)^{-1}(I - B) = \sigma^2 \underbrace{(I - B)(X^T X)^{-1}}_Q$

let $y \neq 0, C \neq 0, d = C^T y$

$Qd = 0$ but $d \neq 0$

$\therefore$ rank-deficient

$$(e) (\hat{C}\beta)^T [C(X^T X)^{-1} C^T]^{-1} (C\beta) \\ = y^T \underbrace{X(X^T X)^{-1} C^T [C(X^T X)^{-1} C^T]^{-1} C(X^T X)^{-1} X^T}_{=A} y$$

$$A^2 = A, \text{rank}(A) = \text{tr}(A) = q$$

$$\frac{(C\hat{\beta})^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta})}{\sigma^2} \sim \chi^2(q, \lambda)$$

$$\text{where } \lambda = \frac{(C\beta)^T [C(X^T X)^{-1} C^T]^{-1} (C\beta)}{2\sigma^2}$$

$$\frac{SSE}{\sigma^2} = \frac{y^T (I - H)y}{\sigma^2} \sim \chi^2(n-p-1)$$

$$A(I - H) = 0$$

$$\Rightarrow F = \frac{(C\hat{\beta})^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta}) / q}{SSE / (n-p-1)} \sim F_{q, n-p-1, \lambda}$$

□

(f)

$$C\beta \in (C\hat{\beta})^T [C(X^T X)^{-1} C^T]^{-1} (C\hat{\beta}) \leq q \cdot s^2 F_{q, n-p-1}(\alpha)$$

$$\text{, where } s^2 = \frac{SSE}{n-p-1}$$

□

2.

(a) Let $Q = A^T A (A^T A)^{-1}$, $P = I$

$$Q A^T A = A^T A, \quad P A^T A = A^T A$$

$$0 = (Q A^T A - P A^T A) (Q^T - P^T)$$

$$= (Q A^T - P A^T) (A Q^T - A P^T)$$

$$= B^T B \Rightarrow B = 0$$

$$A Q^T = A P^T$$

$$\Rightarrow A = A (A^T A)^{-1} A^T A$$

(b) $A A^+ A = A \Rightarrow A^T (A^+)^T A^T = A^T \Rightarrow (A^+)^T = (A^T)^+$

$$\Rightarrow (A (A^T A)^{-1} A^T)^T = A (A^T A)^{-1} A^T$$

$$= A (A^T A)^{-1} A^T \quad ; \text{ symmetric}$$

$$A (A^T A)^{-1} A^T A (A^T A)^{-1} A^T = A (A^T A)^{-1} A^T \quad ; \text{ idempotent}$$

$\underbrace{\hspace{1.5cm}}_{A}$

Let G_1 and G_2 s.t. $A G_1 A^T = A$,
 $A G_2 A^T = A$

$$0 = (A G_1 A^T - A G_2 A^T) (G_1^T A^T - G_2^T A^T)$$

$$= (A G_1 A^T - A G_2 A^T) (A G_1 A^T - A G_2 A^T)^T$$

$$= B^T B \Rightarrow B = 0 \Rightarrow A G_1 A^T = A G_2 A^T ; \text{ invariant}$$

[3].

$$X \sim \begin{cases} F_1 & , u < \frac{1}{3} \\ F_2 & , u \geq \frac{1}{3} \end{cases}$$

$$\begin{aligned} (a) \quad T(F) &= \int b(x) dF(x) \\ &= \int b(x) d\left[\frac{1}{3}F_1 + \frac{2}{3}F_2\right] \\ &= \frac{1}{3} \int b(x) dF_1 + \frac{2}{3} \int b(x) dF_2 \\ &= \frac{1}{3} T(F_1) + \frac{2}{3} T(F_2) \end{aligned}$$

$$\begin{aligned} (b) \quad \int IF(x; T, F_2) dF_1 &= \lim_{\pm \rightarrow 0} \frac{T((1 \pm t)F_2 + \pm F_1) - T(F_2)}{\pm} \\ &= \lim_{\pm \rightarrow 0} \frac{\int b(x) d[(1 \pm t)F_2 + \pm F_1] - \int b(x) dF_2}{\pm} \end{aligned}$$

$$\begin{aligned} &= - \int b(x) dF_2 + \int b(x) dF_1 \\ &= T(F_1) - T(F_2) \end{aligned}$$

$$(c) \quad E_F(X^2) = \int x^2 dF(x) \quad \text{Yes.}$$

$$(d) \quad IF(x; T, F) = x^2 - \int x^2 dF(x)$$

NO. (∵ $E[X^2]$)

4.

$$(a) \frac{\partial ((y - X\beta)^T (y - X\beta) + \lambda \beta^T \beta)}{\partial \beta}$$

$$= -2X^T(y - X\beta) + 2\lambda\beta = 0$$

$$\Rightarrow X^T y - X^T X \beta - \lambda \beta = 0$$

$$\Rightarrow (X^T X + \lambda I) \beta = X^T y$$

$$\Rightarrow \hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

$$= (I + \lambda (X^T X)^{-1})^{-1} \hat{\beta}$$

□

$$(b) \text{LOCV} = \frac{1}{n} \text{PRESS} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_{(-i)})^2$$

$$\hat{\beta}^R = (X^T X + \lambda I_p)^{-1} X^T y$$

$$\hat{\beta}_{(-i)}^R = (X_{(-i)}^T X_{(-i)} + \lambda I_p)^{-1} X_{(-i)}^T y_{(-i)} = (X^T X + \lambda I_p - x_i x_i^T)^{-1} (X^T y - x_i y_i)$$

$$= \left[(X^T X + \lambda I_p)^{-1} + \frac{(X^T X + \lambda I_p)^{-1} x_i x_i^T (X^T X + \lambda I_p)^{-1}}{1 - x_i^T (X^T X + \lambda I_p)^{-1} x_i} \right] (X^T y - x_i y_i)$$

$$= \hat{\beta}^R - \frac{(X^T X + \lambda I_p)^{-1} x_i y_i + \frac{(X^T X + \lambda I_p)^{-1} x_i x_i^T (X^T X + \lambda I_p)^{-1} (X^T y - x_i y_i)}{1 - x_i^T (X^T X + \lambda I_p)^{-1} x_i}}{1 - x_i^T (X^T X + \lambda I_p)^{-1} x_i}$$

$$= \hat{\beta}^R - \frac{(X^T X + \lambda I_p)^{-1} x_i (y_i - \hat{y}_i(\lambda))}{1 - x_i^T (X^T X + \lambda I_p)^{-1} x_i}$$

$$y_i - \hat{y}_{(i)} = y_i - x_i^T \hat{\beta} - \frac{x_i^T (X^T X + \lambda I_p)^{-1} x_i e_i(\lambda)}{1 - x_i^T (X^T X + \lambda I_p)^{-1} x_i}$$

5.

$$(a) l(\beta) = C - \frac{1}{2\sigma^2} (y - X\beta)^T (y - X\beta)$$

$$S(\beta) = \frac{\partial l}{\partial \beta} = \frac{1}{\sigma^2} X^T (y - X\beta) \\ = \frac{1}{\sigma^2} X^T \varepsilon$$

$$\mathbb{E} S(\beta) = 0, \text{var}(S(\beta)) = \frac{1}{\sigma^2} X^T X$$

$$S(\beta) \sim N(0, \sigma^2 X^T X)$$

$$(b) \tilde{\beta}_2 = (X_2^T X_2)^{-1} X_2^T y$$

$$S(0, \tilde{\beta}_2) = \frac{1}{\sigma^2} X_1^T (y - X_2 \tilde{\beta}_2)$$

$$\Rightarrow S_1(0, \tilde{\beta}_2) = \frac{1}{\sigma^2} X_1^T (I - H_2) y$$

$$\mathbb{E} S_1(0, \tilde{\beta}_2) = 0$$

$$\text{var}(S_1(0, \tilde{\beta}_2)) = \frac{1}{\sigma^4} X_1^T (I - H_2) X_1 \sigma^2 = \frac{1}{\sigma^2} X_1^T (I - H_2) X_1$$

$$\varepsilon^T V^{-1} \varepsilon = \sigma^2 S_1(0, \tilde{\beta}_2)^T I \cdot \frac{1}{\sigma^2} X_1^T (I - H_2) X_1^{-1} S_1(0, \tilde{\beta}_2)$$

$$\sim \chi^2(q), \quad \because \text{tr}(I) = q.$$

